

Non- D -finiteness of irreducible NE -prudent ramps

Self-Avoiding Walk Lab

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Publication candidate; priority unresolved; not externally peer reviewed. The theorem below is presented as an internally reviewed mathematical draft. The statement concerns the irreducible NE -prudent-ramp series J only. It makes no claim about the weakly prudent bridge quotient W , the square-lattice connective constant, or publication priority.

Abstract

Let $P(t)$ be the generating function for NE -prudent ramps and let $H(t)$ be the hook-ramp function in the explicit algebraic formulas of Bacher and Beaton. Their irreducible NE -prudent-ramp series is

$$J(t) = \frac{P(t) - H(t)}{1 + P(t)}.$$

We prove that J is not D -finite. The proof starts from the exact kernel expressions and their meromorphic continuation. A change of parameter $q = U(t, 1) = \exp(-\varepsilon/2)$ isolates a critical crossing in the iterated product defining P and H . The product is factored exactly into a smooth factor and a Gamma ratio. Uniform Gamma estimates, an integrable bound at the crossing, and exponential control of the far tail give uniform limits for two regularized moments. Their limits are four one-dimensional integrals with an explicit singular profile. Outward dyadic interval arithmetic certifies endpoint signs for the limiting functions and a positive hook margin throughout a phase interval. For all sufficiently large integer indices, continuity then gives a zero of $1 + P$ with $1 + H > 0$. These zeros yield infinitely many non-removable poles of J , which is impossible for a D -finite analytic germ.

Keywords: self-avoiding walks; prudent walks; generating functions; D -finite series; kernel method; Gamma products.

1 Introduction

Prudent self-avoiding walks are walks on the square lattice that never step towards a previously visited vertex. They form several nested subclasses whose generating functions exhibit a range of behaviours between directed models and unrestricted self-avoiding walks. We use the following precise convention. A walk $\omega = (\omega_0, \dots, \omega_n)$ is rooted at the origin, and its bounding box is the smallest axis-parallel rectangle containing its vertices. A walk is prudent when no step points towards a previously visited vertex, and it is NE -prudent when it is prudent and, for every prefix, the prefix endpoint lies on the north or east side of that prefix's bounding box. It is a bridge when, writing $y_i = y(\omega_i)$, $y_0 < y_i \leq y_n$ for every $i \geq 1$. A bridge is a ramp when $x(\omega_i) \leq x(\omega_n)$ for every $0 \leq i \leq n$; throughout, ramps are nonempty bridges. Bridge concatenation is the usual concatenation at the endpoint; “irreducible” means that no factorization into two shorter bridges exists. All generating functions below weight a walk by t^n , with no quotient by lattice symmetries.

In the model studied here, Bacher and Beaton introduced weakly prudent bridges and expressed the irreducible NE -prudent-ramp series in terms of two auxiliary functions. Their Theorem 17 gives the meromorphic continuation and the infinitely many source poles accumulating at $\sigma = \sqrt{2} - 1$ [1]; see also the [official PDF](#). The quotient defining the irreducible series requires a separate noncancellation argument.

We use the notation

$$J(t) = \frac{P(t) - H(t)}{1 + P(t)}.$$

Here P counts the NE -prudent ramps and H is the hook-ramp function denoted $\tilde{P}$ in the source. The common poles of P and H do not by themselves survive in this quotient: at a common simple pole the local principal parts can cancel after the quotient is formed. Our argument instead finds infinitely many zeros of $1 + P$ between successive source poles and proves that the numerator $-1 - H$ is nonzero at those zeros.

Theorem 1.1 (Main result). *The formal power series at the origin represented by*

$$J(t) = \frac{P(t) - H(t)}{1 + P(t)}$$

is not D -finite.

The proof is conditional only on the exact formulas and meromorphic continuation theorem established in [1]; all additional limits and signs needed for the quotient are proved below. The interval calculation used to evaluate the limiting integrals is described explicitly, including its endpoint substitutions and outward rounding. It does not use floating-point quadrature error as a proof.

Our conclusion is deliberately narrow. We do not transfer it to the additional weakly prudent bridge series W , do not state a formula for any square-lattice connective constant, and do not make a claim of being first.

2 The exact Bacher–Beaton functions

We first record the formulas used in the proof. This fixes the branch, normalization, and all auxiliary quantities before any limiting operation. For t and v near the origin, set

$$\alpha(t, v) = 1 - tv + t^2 + t^3v$$

and let

$$U(t, v) = \frac{\alpha(t, v) - \sqrt{\alpha(t, v)^2 - 4t^2}}{2t}. \tag{1}$$

The square root is the branch for which $U(t, v) = t + O(t^2)$. Put

$$q = U(t, 1), \quad u_n = U(t, q^{2n}), \quad w_n = U(t, q^{2n+2}).$$

For the real parameter range used later, all of these quantities are on the same physical branch.

Write

$$C = q - t, \quad D = 1 - tq, \quad E = 1 - q^2, \quad h(u) = tq - Cu.$$

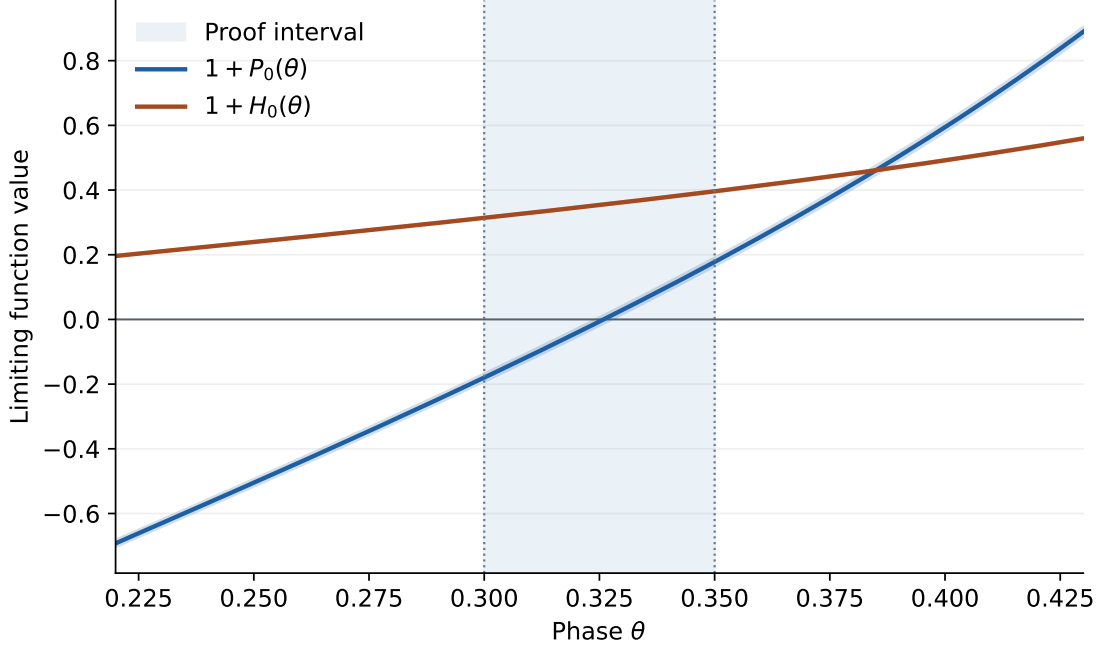


Figure 1: Illustrative limiting phase plot for $1 + P_0$ and $1 + H_0$. The shaded interval is $\theta \in [0.30, 0.35]$, and the bands visualize the certified integral coefficient intervals; the exact sign conclusions are the interval statements in Section 8.

For u, w on the physical branch, define

$$\mathcal{A}(t; u, w) = \frac{C(1 - t^2)(Du - [qC + tEu]w)}{D(1 - tu)(1 - tw)h(u)}, \quad (2)$$

$$\mathcal{B}(t; u, w) = \frac{qC^2(t - Dw)}{D^2h(u)}, \quad (3)$$

$$\begin{aligned} \tilde{\mathcal{A}}(t; u, w) &= \frac{t^2(1 - t^2)C}{D(1 - tu)^2(1 - tw)^2h(u)} \\ &\quad \times [Du^2(1 - 2tw) - \{qC - tu(2qC + tEu)\}w^2]. \end{aligned} \quad (4)$$

Let $p_0 = 1$ and

$$p_{n+1} = p_n \mathcal{B}(t; u_n, w_n).$$

Then the two functions from Propositions 14 and 16 of [1] are

$$P(t) = \frac{q(1 - t^2)}{D} + q \sum_{n \geq 0} p_n \mathcal{A}(t; u_n, w_n), \quad (5)$$

$$H(t) = \frac{t^2 q^2 (1 - t^2)}{D^2} + q \sum_{n \geq 0} p_n \tilde{\mathcal{A}}(t; u_n, w_n). \quad (6)$$

For fixed t away from the source poles, these sums converge in the meromorphic continuation supplied by the source. The series at the origin is the germ to which the continuation is attached.

The kernel equation at $v = 1$ gives

$$q + \frac{1}{q} = \frac{1}{t} - 1 + t + t^2. \quad (7)$$

At $t = \sigma = \sqrt{2} - 1$, the right side equals 2, and its derivative with respect to t is nonzero. Thus the smaller branch $q < 1$ can be used as a local parameter near σ .

The source proves in Theorem 17 that P and H have simple real poles $a_0 < a_1 < \dots < \sigma$, with $a_\ell \rightarrow \sigma$, and no other poles between consecutive members of this sequence. In the notation above,

$$h(u_n) = 0 \iff (q - t)(u_n - t) = t^2 \iff \frac{(q - t)(U(t, q^{2n}) - t)}{t^2} = 1. \quad (8)$$

Thus the zeros of the displayed denominator are exactly the pole characterization used in that theorem. We use this continuation theorem to evaluate the formulas on the intervals constructed below.

3 Exact algebra and regularized moments

The critical argument is easier after a common algebraic reduction of the two summands. Define

$$g(u) = t - Du, \quad f(u) = \frac{u}{1 - tu}, \quad k = \frac{qC^2}{D^2}.$$

Two identities following from (7) are

$$CD = tq(1 - t^2), \quad h(u) - kg(u) = \frac{CE}{D}(1 - u). \quad (9)$$

For $p \geq 0$, put

$$\mathcal{A}_p(t; u, w) = \frac{C(1 - t^2)}{Dh(u)}(Df(u)^p - qCf(w)^p).$$

Direct expansion of (2) and (4) gives

$$\mathcal{A} = \mathcal{A}_1, \quad \tilde{\mathcal{A}} = t^2 \mathcal{A}_2. \quad (10)$$

For example,

$$Df(u) - qCf(w) = \frac{Du - [qC + tEu]w}{(1 - tu)(1 - tw)},$$

which proves the first equality. For the hook term, the numerator identity

$$\begin{aligned} Df(u)^2 - qCf(w)^2 &= \frac{Du^2(1 - 2tw)}{(1 - tu)^2(1 - tw)^2} \\ &\quad - \frac{\{qC - tu(2qC + tEu)\}w^2}{(1 - tu)^2(1 - tw)^2} \end{aligned}$$

proves the second.

Let

$$g_n = g(u_n), \quad h_n = h(u_n), \quad z_n = \frac{p_n}{g_n}.$$

The product recurrence becomes

$$z_{n+1} = z_n \frac{qC}{D} \frac{g_n}{g_n + \delta}, \quad \delta = \frac{t^2 E}{C}. \quad (11)$$

The following telescoping identity is exact:

$$p_n \mathcal{A}_p(t; u_n, w_n) = Y_{p,n} - Y_{p,n+1} - t^2 E(1 - t^2) \frac{p_n f(u_n)^p}{h_n g_n}, \quad (12)$$

where

$$Y_{p,n} = p_n \frac{D(1 - t^2) f(u_n)^p}{g_n}.$$

Indeed, substitution of (10) into the right side reduces (12) to $Dh(u) - t^2 E = Cg(u)$, which is equivalent to (9). The fixed- t tail is locally geometric, so summing (12) gives the two exact moment representations

$$P(t) = \frac{C}{g(q)} + \sum_{n \geq 0} W_n f(u_n), \quad (13)$$

$$H(t) = t^2 f(q) \frac{C}{g(q)} + t^2 \sum_{n \geq 0} W_n f(u_n)^2, \quad (14)$$

where

$$W_n = -qt^2 E(1 - t^2) \frac{p_n}{h_n g_n}. \quad (15)$$

The same algebra gives the exact zeroth-moment relation

$$\sum_{n \geq 0} W_n (1 - u_n) = -\frac{tD^2}{g(q)}. \quad (16)$$

This relation is signed; it is not a positivity estimate.

The summands in (13)–(14) have apparent poles at the zeros of h . We remove them before taking the critical limit. Let

$$u_h = \frac{tq}{C}, \quad h(u_h) = 0, \quad L_j = \frac{f(u_h)^j}{1 - u_h},$$

and

$$R_j(u) = f(u)^j - L_j(1 - u), \quad j = 1, 2.$$

Since $h(u) = -C(u - u_h)$,

$$-\frac{R_j(u)}{h(u)} = \frac{1}{C} \left[\frac{f(u)^j - f(u_h)^j}{u - u_h} + L_j \right], \quad (17)$$

with the diagonal value interpreted by continuity. The right side is uniformly bounded on the physical interval for t near σ . Writing

$$Q = qt^2 E(1 - t^2), \quad S_0 = -\frac{tD^2}{g(q)},$$

the exact regularized moments are

$$P(t) = \frac{C}{g(q)} + L_1 S_0 - Q \sum_{n \geq 0} z_n \frac{R_1(u_n)}{h_n}, \quad (18)$$

$$\frac{H(t)}{t^2} = f(q) \frac{C}{g(q)} + L_2 S_0 - Q \sum_{n \geq 0} z_n \frac{R_2(u_n)}{h_n}. \quad (19)$$

The cancellation in (17) is exact at finite t ; no divergent moments are subtracted only after taking a limit.

4 Critical parameter and phase

Set

$$\varepsilon = -\log(q^2), \quad q = e^{-\varepsilon/2}.$$

Equation (7) and the implicit function theorem give a real analytic function $t = t(\varepsilon)$ near $\varepsilon = 0$, with

$$t(0) = \sigma, \quad t(\varepsilon) < \sigma \quad (\varepsilon > 0 \text{ small}).$$

For $s \geq 0$, define

$$g_\varepsilon(s) = t - DU(t, e^{-s}).$$

The function g_ε has a simple positive zero $s_g(\varepsilon)$. The zero of $g_\varepsilon(s) + \delta$ is denoted by $s_h(\varepsilon)$. At the critical point,

$$s_g(\varepsilon) \longrightarrow s_* = -\log\left(\frac{1}{2} + \frac{\sqrt{2}}{4}\right), \quad U(\sigma, e^{-s_*}) = u_* = \frac{1}{\sqrt{2}}.$$

The local algebraic kernel has no branch singularity at this positive crossing. Differentiating the kernel relation at the crossing gives

$$\frac{s_g - s_h}{\varepsilon} \longrightarrow \eta = \frac{1}{\sqrt{2}}, \quad \frac{-\log(qC/D)}{\varepsilon} \longrightarrow \lambda = \frac{1}{1 - \sigma}.$$

In particular, $\eta < 1$, the fact that will make the crossing integrable.

Define

$$a = \frac{s_g}{\varepsilon}, \quad b = \frac{s_h}{\varepsilon}.$$

Then $b = a - \eta_\varepsilon$, where $\eta_\varepsilon = \eta + O(\varepsilon)$. Fix the compact phase interval

$$\mathcal{T} = \left[\frac{3}{10}, \frac{7}{20} \right].$$

Since $s_g(\varepsilon) = s_* + O(\varepsilon)$, one has

$$\frac{da}{d\varepsilon} = -\frac{s_*}{\varepsilon^2} + O(1).$$

Therefore, for every sufficiently large integer N and every $\theta \in \mathcal{T}$, there is a unique small $\varepsilon_N(\theta)$ such that

$$a(\varepsilon_N(\theta)) = N + \theta. \tag{20}$$

The inverse is continuous in θ , and $\varepsilon_N(\theta) = s_*/N + O(N^{-2})$ uniformly on $\mathcal{T}$. The separation of $\mathcal{T}$ from the integers, together with $\eta = 1/\sqrt{2}$, implies that a and b remain a fixed positive distance from all integers. Thus no $g_\varepsilon(n\varepsilon)$ or $g_\varepsilon(n\varepsilon) + \delta$ vanishes in the phase family.

5 Matched Gamma products

The product in (11) is singular near the crossing, so it must not be replaced there by an Euler approximation. We factor its singular part exactly. For a fixed $S > s_* + 1$, set

$$F_\varepsilon(s) = \frac{g_\varepsilon(s)}{s - s_g}, \quad G_\varepsilon(s) = \frac{g_\varepsilon(s) + \delta}{s - s_h}, \quad R_\varepsilon(s) = \frac{F_\varepsilon(s)}{G_\varepsilon(s)}.$$

The removable values at the two zeros are filled in by the corresponding positive derivatives. At $s_j = j\varepsilon$,

$$\frac{g_\varepsilon(s_j)}{g_\varepsilon(s_j) + \delta} = \frac{j-a}{j-b} R_\varepsilon(s_j).$$

Consequently, with $r = qC/D$,

$$z_n = z_0 r^n T_n(a, b) \prod_{j=0}^{n-1} R_\varepsilon(j\varepsilon), \quad T_n(a, b) = \prod_{j=0}^{n-1} \frac{j-a}{j-b}. \quad (21)$$

The singular product has the exact Gamma representation

$$T_n(a, b) = \frac{\Gamma(n-a)\Gamma(-b)}{\Gamma(-a)\Gamma(n-b)}. \quad (22)$$

All products at $n = 0$ are one.

The smooth factor has a uniform limit. On $0 \leq s \leq S$,

$$\frac{\log R_\varepsilon(s)}{\varepsilon} \longrightarrow \psi(s) = -\frac{\kappa}{g_0(s)} + \frac{\eta}{s-s_*}, \quad \kappa = \frac{\sigma^2}{1-\sigma},$$

where the apparent pole at s_* is removable. To prove this, take a fixed neighborhood of s_* . There g_ε is analytic with a uniformly simple zero, and Taylor expansion in the two root locations gives the uniform first-order expansion of the ratio of divided differences. Outside that neighborhood,

$$\log R_\varepsilon(s) = \log \left(1 + \frac{s_g - s_h}{s - s_g} \right) - \log \left(1 + \frac{\delta}{g_\varepsilon(s)} \right),$$

and both terms have uniformly bounded denominators and are $O(\varepsilon)$. The explicit square-root expression (1) gives uniform convergence of g_ε on the whole compact interval, including $s = 0$. The two local descriptions agree at s_* , so the Riemann-sum limit is uniform:

$$\sum_{j < n} \log R_\varepsilon(j\varepsilon) \longrightarrow \int_0^s \psi(x) dx, \quad n\varepsilon \longrightarrow s. \quad (23)$$

Likewise $n \log r \rightarrow -\lambda s$, uniformly on bounded s -intervals.

For the Gamma factor, use the reflection formula and the uniform parameter version of the Gamma-ratio asymptotic $\Gamma(x+c)/\Gamma(x+d) \sim x^{c-d}$. These are the standard identities [3, Eq. 5.5.3] and [2, Eq. 5.11.12]; the exact source pages are [DLMF 5.5.3](#) and [DLMF 5.11.12](#). If $n\varepsilon \rightarrow s$ and $s \neq s_*$, then

$$T_n(a, b) \longrightarrow \begin{cases} \left(\frac{s_*}{|s-s_*|} \right)^\eta, & s < s_*, \\ \frac{\sin(\pi\theta)}{\sin(\pi(\theta-\eta))} \left(\frac{s_*}{|s-s_*|} \right)^\eta, & s > s_*. \end{cases} \quad (24)$$

The rate $\eta_\varepsilon - \eta = O(\varepsilon)$ is essential: it implies $(\eta_\varepsilon - \eta) \log \varepsilon \rightarrow 0$, so no unnoticed power of ε remains. The convergence is uniform on compact s -subintervals avoiding s_* and uniformly in $\theta \in \mathcal{T}$.

We also need a bound through the crossing. For $n \leq N$, reflection is not needed and the positive-argument identity is

$$T_n = \frac{\Gamma(a+1)\Gamma(b-n+1)}{\Gamma(b+1)\Gamma(a-n+1)}.$$

For $n \geq N + 1$, reflection gives

$$T_n = \frac{\sin(\pi a) \Gamma(a+1) \Gamma(n-a)}{\sin(\pi b) \Gamma(b+1) \Gamma(n-b)}.$$

All Gamma arguments that can be small remain a fixed distance from zero because of the phase restrictions. Uniform positive-argument Gamma-ratio bounds therefore give, for any fixed $\nu > 0$ with $\eta + \nu < 1$,

$$|z_n| \leq C_S(\varepsilon + |n\varepsilon - s_g|)^{-\eta-\nu}, \quad n\varepsilon \leq S, \quad (25)$$

with C_S independent of N and θ . The smooth product and r^n are bounded on this compact scale. Beyond a fixed $S > s_* + 1$, g_ε is uniformly positive and

$$0 < \frac{g_\varepsilon(s)}{g_\varepsilon(s) + \delta} \leq 1, \quad r \leq e^{-c\varepsilon}$$

for some $c > 0$. The recurrence then gives

$$|z_n| \leq C e^{-c(n\varepsilon - S)}, \quad n\varepsilon \geq S. \quad (26)$$

The first index beyond S is controlled by (25). This proves the uniform product and tail bounds required below.

6 The limiting product profile

At criticality write

$$u = U(\sigma, e^{-s}), \quad d = 1 - \sigma, \quad u_* = \frac{1}{\sqrt{2}}.$$

The inverse relation between s and u is

$$e^{-s} = \phi(u) = \frac{(u - \sigma)(1 - \sigma u)}{\sigma(1 - \sigma^2)u}. \quad (27)$$

The limiting logarithmic derivative obtained from (23), (24), and $n \log r$ is

$$\frac{d}{ds} \log |Z(u(s))| = -\lambda - \frac{\kappa}{g_0(s)}. \quad (28)$$

Substitution of (27) into the right-hand side and elementary partial fractions verify the explicit profile below on each side of u_* . Normalize the positive profile by

$$M_{\text{raw}}(u) = (u - \sigma)^{2+\sqrt{2}} \left(\frac{1}{\sigma} - u \right)^{\sqrt{2}} |u - u_*|^{-\eta} u^{-1-\sqrt{2}},$$

$$M(u) = \frac{M_{\text{raw}}(u)}{M_{\text{raw}}(1)}.$$

Let

$$A_* = \frac{1}{\sigma^2}, \quad B(\theta) = A_* \frac{\sin(\pi\theta)}{\sin(\pi(\eta - \theta))}.$$

For $\theta \in \mathcal{T}$, $B(\theta) > 0$. Combining (21), (23), and (24), while using $g_0(0) = -\sigma^2$, gives the two-sided limiting product

$$Z_\theta(u) = \begin{cases} -A_* M(u), & u_* < u \leq 1, \\ B(\theta) M(u), & \sigma < u < u_*. \end{cases} \quad (29)$$

The relative amplitude in the second line is not a free choice: it is the reflection multiplier in (24). This is the connection information that would be lost by solving only the formal differential equation for the smooth product.

7 Uniform convergence of the regularized moments

At the critical point let

$$f(u) = \frac{u}{1 - \sigma u}, \quad L = 2 + \sqrt{2}, \quad K = \sigma^2(1 - \sigma^2).$$

The limits of the finite- ε regularizers in (17) are

$$\begin{aligned} V_1(u) &= \frac{1}{d} \left[L + \frac{1}{u_*(1 - \sigma u)} \right], \\ V_2(u) &= \frac{1}{d} \left[L + \frac{f(u) + 1}{u_*(1 - \sigma u)} \right]. \end{aligned}$$

Both are continuous and positive on $[\sigma, 1]$. The boundary terms in (18)–(19) have the limits $-d$ and -3 , respectively, after the factor t^2 is removed from the second equation. This follows by substituting $t \rightarrow \sigma$, $q \rightarrow 1$, and (9); also $Q/\varepsilon \rightarrow K$.

Define four finite positive integrals:

$$J_{j,\text{pre}} = K \int_{u_*}^1 M(u) V_j(u) \frac{\phi'(u)}{\phi(u)} du, \quad (30)$$

$$J_{j,\text{post}} = K \int_{\sigma}^{u_*} M(u) V_j(u) \frac{\phi'(u)}{\phi(u)} du, \quad j = 1, 2. \quad (31)$$

The only interior singularity is $|u - u_*|^{-\eta}$, with $\eta < 1$, so both integrals converge. Uniformly for $\theta \in \mathcal{T}$, the regularized moment sums converge to

$$P_0(\theta) = -d - A_* J_{1,\text{pre}} + B(\theta) J_{1,\text{post}}, \quad (32)$$

$$H_0(\theta) = \sigma^2 [-3 - A_* J_{2,\text{pre}} + B(\theta) J_{2,\text{post}}]. \quad (33)$$

Proof of the uniform moment convergence. Fix a small $\nu > 0$ with $\eta + \nu < 1$, and a crossing window $|n\varepsilon - s_g| < \rho$. The divided-difference formula (17) gives a uniform bound for the regularized factor $R_j(u_n)/h_n$. Hence (25) bounds the absolute contribution of the crossing window by

$$C\varepsilon \sum_{|n\varepsilon - s_g| < \rho} (\varepsilon + |n\varepsilon - s_g|)^{-\eta - \nu} \leq C'(\rho + \varepsilon)^{1 - \eta - \nu}.$$

This tends to zero uniformly when first $\varepsilon \rightarrow 0$ and then $\rho \rightarrow 0$. There is no residual point mass at the crossing.

Outside this window, the product convergence (23) and (24) are uniform, so ordinary Riemann-sum convergence gives (30)–(33). The endpoint $s = 0$ is covered by (25); the square-root behaviour of the algebraic kernel there is not differentiated in the argument. Finally, (26) gives a uniform exponentially small bound beyond the fixed S . Letting $S \rightarrow \infty$ completes the proof uniformly in θ . $\square$

8 Certified signs of the limiting functions

We now describe the exact sign certificate for (32) and (33). It is included to distinguish a rigorous interval calculation from a floating-point quadrature.

For the pre-integrals in (30), substitute

$$u = u_* + (1 - u_*)x^8, \quad 0 \leq x \leq 1.$$

For the post-integrals, use

$$u = u_* - (u_* - \sigma)x^8, \quad 0 \leq x \leq 1,$$

with the orientation reversed so that the integral runs from σ to u_* . In both cases

$$|u - u_*|^{-\eta} |du| = 8\Delta^{1-\eta} x^{8(1-\eta)-1} dx,$$

where Δ is the corresponding interval length. The exponent $8(1-\eta) - 1$ is strictly positive.

The certificate partitions $[0, 1]$ into 2048 equal bins. It encloses every nonnegative base and exponent by dyadic intervals, evaluates sixteenth-bit exponent bounds using 16 outward square roots and integer powers, and takes the hull when a base interval straddles one. Endpoint factors are algebraically cancelled before interval evaluation. The constant π is enclosed with Machin's identity and alternating arctangent remainders; the sine factor in $B(\theta)$ is enclosed by a Taylor polynomial with its Lagrange remainder. All arithmetic in this calculation is rational outward interval arithmetic.

The resulting decimal displays of the certified integral intervals are

	J_1	J_2
pre	[0.45986641, 0.46232754]	[0.650478, 0.65423355]
post	[0.42401814, 0.42637240]	[0.56790451, 0.57100730].

Substitution into the limiting functions gives

θ	$1 + P_0(\theta)$	$1 + H_0(\theta)$
3/10	[-0.19278831, -0.16685274]	[0.31077599, 0.31715257]
7/20	[0.16373676, 0.19165184]	[0.39270347, 0.39952767].

The displayed intervals are rounded outward from the exact dyadic enclosures.

The hook margin holds throughout the phase interval, not only at its endpoints. Indeed,

$$B'(\theta) = A_* \frac{\pi \sin(\pi\eta)}{\sin^2(\pi(\eta - \theta))} > 0$$

on $\mathcal{T}$, and $J_{2,\text{post}} > 0$. Thus $H_0(\theta) + 1$ is increasing, and

$$H_0(\theta) + 1 > 0.310 \quad (\theta \in \mathcal{T}). \quad (34)$$

9 Infinitely many non-removable poles

We now prove the main theorem. For each sufficiently large N , define

$$t_N(\theta) = t(\varepsilon_N(\theta)), \quad \theta \in \mathcal{T},$$

using (20). Uniform convergence in (32)–(33) and the strict certified margins imply

$$P(t_N(3/10)) < -1 < P(t_N(7/20)), \quad H(t_N(\theta)) + 1 > 0 \quad (\theta \in \mathcal{T})$$

for all sufficiently large N . The source pole sequence is avoided throughout these phase intervals by the distance conditions on a and b , so P and H are analytic there. Continuity gives a point r_N in the interval with

$$P(r_N) = -1.$$

At this point,

$$P(r_N) - H(r_N) = -1 - H(r_N) \neq 0.$$

Consequently J has a non-removable pole at r_N . The intervals for different N are disjoint because their phase parameters satisfy $a = N + \theta$, and $r_N \rightarrow \sigma$. We have therefore produced infinitely many distinct finite-plane poles of the analytic continuation of J .

For completeness, recall the elementary D -finite obstruction. If a nonzero analytic germ at the origin is D -finite, then it satisfies a linear differential equation with polynomial coefficients. Its analytic continuation can have finite singularities only at zeros of the leading coefficient of that equation, a finite set. The infinitely many poles just constructed contradict this property. Theorem 1.1 follows. $\square$

10 Scope and reproducibility

The exact starting formulas (5)–(6), the algebraic identities, the product factorization, the uniform bounds, and the limiting integral signs are separate proof components. The interval certificate described in Section 8 is a finite exact-arithmetic certificate for the displayed limiting integrals; the passage from the discrete products to those integrals is supplied by Sections 5–7.

The proof establishes non- D -finiteness only for J . It does not establish the corresponding claim for the weakly prudent bridge quotient W , and it does not assert an exact growth constant for unrestricted self-avoiding walks. The priority of the result is unresolved and the draft has not been externally peer reviewed.

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